

PAPER_018: Aether Noise Spectrum Characterization for LISA

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Authors: Daniel Murphy & UQFF Research Collective **Date:** 2026-03-07 **Domain:** 1.2 — Gravitational Waves — LISA Future Detector **Primary Validation File:** validate_lisa_extended.py **Calibration constants:** $\kappa = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$

Abstract

We characterize the spectral signature of UQFF aether fields in the LISA frequency band (0.1–100 mHz). UQFF predicts that *Um vacuum field oscillations imprint narrow spectral lines at harmonics of $f_U \approx 1$ mHz* onto the stochastic gravitational wave background (SGWB), with an integrated aether power fraction of 222.93% relative to the GR SGWB. The peak aether feature at $f_{\text{peak}} = 0.99$ mHz yields an integrated detection SNR of 12,695,834—trivially detectable by LISA over a 4-year mission. A broad TRZ suppression dip ($\sim 10\%$) appears near 5 mHz. These spectral features have no astrophysical analogue and constitute a smoking-gun test of UQFF vacuum structure.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Motivation

UQFF predicts that the vacuum supports actively coupled aether fields characterized by:

U_m: Magnetic energy parameter (= 1.0 in calibrated UQFF)

β_m : Modulation parameter (= 0.01)

f_TRZ: Trans-zero frequency factor (= 0.1)

These fields interact with propagating GWs to create:

Spectral **power excess** at fundamental and harmonic frequencies

Sideband pairs from β_m modulation

A broad **suppression dip** from TRZ near 5 mHz

2. Aether Noise Spectrum Model

The total stochastic power spectral density (PSD) in the LISA band:

$$S_{UQFF}(f) = S_{GR}(f) \times [1 + P_{aether}(f)] \times F_{TRZ}(f)$$

where the GR background follows $S_{GR}(f) \propto \Omega_{GW}(f) \propto f^{2/3}$ (inspiral-dominated), and:

$$S_{\{UQFF\}}(f) = S_{\{GR\}}(f) \cdot [1 + P_{\{aether\}}(f)] \cdot F_{\{TRZ\}}(f)$$

$$P_{\{aether\}}(f) = U_m \sum_n e^{-n/2} \cdot \delta(f - n f_U) \cdot W(\Delta f)$$

$$F_{\{TRZ\}}(f) = 1 - 0.1 \cdot \exp\left[-\frac{(f - f_{\{TRZ\}})^2}{2\sigma_{\{TRZ\}}^2}\right]$$

Key numerical results: $U_m = 1.0e-4$, $f_U(\text{harmonic } 1) = 9.9e-1 \text{ mHz} = 9.9e-4 \text{ Hz}$, $\Omega_{GW} \sim 1.0e-9$, F_{TRZ} peak suppression = $1.0e-1$ (10%)

$$Paether(f) = U_m \times \sum_n \exp(-n/2) \times \delta(f - n f_U) \times W(\Delta f)$$

with spectral line width $W(\Delta f) \approx 10\%$ fractional bandwidth, and:

$$FTRZ(f) = 1 - 0.1 \times \exp[-(f - f_{TRZ,peak})^2 / (2\sigma_{TRZ}^2)]$$

3. Spectral Components

Component	Frequency	Amplitude	Origin
GR SGWB	0.1–100 mHz	$\Omega_{GW} \sim 10^{-9}$	Inspiral population
U_m harmonic n=1	0.99 mHz	$U_m \times 1.0$	Fundamental aether line
U_m harmonic n=2	~2 mHz	$U_m \times e^{-1}$	1st harmonic
U_m harmonic n=3	~3 mHz	$U_m \times e^{-1} \cdot 1/2$	2nd harmonic
U_m harmonics n=4,5	~4, 5 mHz	$U_m \times e^{-1/2}$	Higher harmonics
β_m sidebands	$f_n \pm 0.01 \text{ mHz}$	$\beta_m \times \text{amplitude}$	Modulation pairs
TRZ suppression	~5 mHz	−10%	Trans-zero dip

4. Quantitative Results

Metric	Value
Frequency range	0.10 – 100 mHz
Spectral bins	200
Spectral resolution Δf	$8 \times 10^{-4} \text{ mHz}$
Observation time	4 years
Aether power fraction $Paether/PGR$	222.93%
Peak aether frequency f_{peak}	0.99 mHz
Integrated detection SNR	12,695,834
Detectable (SNR > 5)	True

The 222.93% aether power fraction means the UQFF aether signal is more than **twice** the GR SGWB in the LISA band—an unmistakably strong signature.

5. Comparison With Known Noise Sources

Source	Spectrum Shape	Prediction	UQFF Distinguishable?
GR SGWB (inspiral)	$\propto f^{(2/3)}$	$\Omega_{GW} \sim 10^{-9}$	Baseline
Galactic WD foreground	Peaked at ~3 mHz	Reduced by UQFF factor	Yes — WD band modified

Source	Spectrum Shape	Prediction	UQFF Distinguishable?
LISA instrument noise	$\propto f^2$ (low f)	Not modified	Yes — different spectral shape
Cosmological SGWB	Flat (inflation)	Not modified	Yes — UQFF adds lines
UQFF aether lines	Harmonic comb	222.93%	Unique signature

6. Validation

Run command (Windows-safe):

```
set PYTHONUTF8=1
python validate_lisa_extended.py
```

Test status from `compute_aether_noise_spectrum()`:

Check	Result
$P_{\text{aether}}/P_{\text{GR}} > 100\%$	■ PASS (222.93%)
f_{peak} identified	■ PASS (0.99 mHz)
Integrated SNR > 5	■ PASS (12,695,834)
Detectable = True	■ PASS

TEST STATUS: PASSED: `compute_aether_noise_spectrum`

7. Observational Strategy for LISA

Targeted narrow-band search at 1, 2, 3 mHz: Comb filter for U_m harmonic signature with 10% bandwidth windows

Cross-correlation test: Compare LISA data-stream with predicted sideband pattern at $f_n \pm f_{\text{mod}}$ ($f_{\text{mod}} = 0.01$ mHz)

TRZ notch identification: Low-pass filter around 5 mHz to identify broad suppression against WD foreground

Time-series modulation: Look for ~10% amplitude oscillations at ~1-hour period in GW background estimation

8. Conclusion

UQFF aether fields create a distinctive spectral fingerprint in the LISA band: a harmonic comb at multiples of ~1 mHz with 222.93% integrated power relative to the GR background, and a ~10% TRZ suppression near 5 mHz. With detection SNR > 12 million, these signatures are trivially observable by LISA and have no counterpart in standard astrophysics. A single 4-year LISA dataset is sufficient to confirm or rule out UQFF vacuum field coupling at high significance.

Validator: `validate_lisa_extended.py` — PASSED (`computeaethernoise_spectrum`)

PASSED: `computeaethernoise_spectrum`

ALL TESTS PASSED - LISA extended methods validated

References

validate_lisa_extended.py

paper18_validate_lisa_extended.out